

CHOROIDAL MELANOMA

Does Endoresection Prevent Neovascular Glaucoma in Patient Treated with Proton Beam Irradiation?

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Purpose: To evaluate the efficacy of endoresection after proton beam radiotherapy to prevent neovascular glaucoma (NVG) in patients treated for choroidal melanoma.

Methods: From a series of 4,867 patients treated for choroidal melanoma were prospectively recorded in the database (Macro Inferred 3.075). One hundred and seventy-one patients presenting a tumor diameter >10 mm and thickness >5 mm treated with proton beam (PB) radiotherapy were selected. One group of 63 patients was treated with PB therapy followed by endoresection (PE) of the scar. This group was compared with 2 historical matched controlled groups: 57 patients treated with PB therapy alone (P) and 51 patients treated with PB therapy followed by transpupillary thermotherapy of the scar (PTTT). Main outcome measures are as follows: age, gender, tumor diameter, tumor thickness, pre- and posttreatment visual acuity, NVG rate, secondary enucleation rate, and 5-year survival. Statistical analysis was performed using R version 2.5.1 software.

Results: Correlations between the 3 groups were $P = 0.29$ for age, $P = 4.7 \times 10^{-12}$ for tumor diameter, and $P = 6.44 \times 10^{-4}$ for tumor thickness. Comparison between the 3 groups showed that 2-year survival without secondary enucleation was 96.2% for PE, 88.8% for P, and 98% for PTTT ($P = 0.203$) (95% confidence interval). Two-year survival without NVG (95% confidence interval) was 92.7% (85.1–1.00) for PE, 54.6% for P, and 62.1% for PTTT ($P = 0.0001$). The difference between the endoresection (PE) group and the PB radiotherapy (P) and PB radiotherapy + TTT (PTTT) groups in terms of reduction of the NVG rate was statistically significant. Relative risk of developing NVG was calculated with the P group as reference, relative risk = 1. The relative risk of the PTTT group was 0.79 (20% reduction of the risk), and the relative risk of the PE group was 0.18 (82% reduction of the risk of developing NVG).

Conclusion: This study shows that endoresection of the necrotic scar after PB radiotherapy reduces the risk of NVG and secondary enucleation for selected choroidal melanoma patients.

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Treatment of uveal melanoma is evolving continuously for patient benefits. Conservative therapy using different methods of irradiation (brachytherapy or teletherapy) has been used for many years with satisfying results in terms of tumor control and eye retention.^{1,2} There is no ideal method to conserve the eye for the largest tumors. Side effects of irradiation whatever the method used are well known. Some severe complications leading to pain, blindness, and secondary enucleation are exudative retinal detachment

and neovascular glaucoma (NVG).³ Studies have shown that for tumors of >15 mm in diameter and more than 8 mm in thickness, the risk of complications at 5 years is of 52% for NVG and 27% for retinal detachment.^{4–6} Treatment of the scar with laser or surgical exoresection or endoresection seems efficient in reducing the rate of complications.^{7–10} However, the efficiency to prevent NVG by endoresection still needs to be demonstrated and the best indications need to be found.

We studied the results in term of NVG reduction in patients treated with proton beam radiotherapy combined with endoresection of the scar compared with two historical matched controlled groups of patients. The first one was treated with proton beam radiotherapy alone, and the second one was treated with proton beam radiotherapy combined with transpupillary thermotherapy (TTT) of the scar. These three groups were then compared with determine if endoresection of the scar reduces significantly the risk of NVG and/or secondary enucleation.

Patients and Methods

Retrospective Observational Case Series

From January 1990 to April 2011, 4,867 patients treated at Institut Curie for choroidal melanoma were recorded in our clinical trial software Macro Infermed 3.075 (InferMed U.K.) database; 3,374 patients of them were treated by proton beam radiotherapy.¹¹

Patients were treated with proton beam with the same protocol. Patients were treated by proton beam with the same protocol. Proton beam was delivered by a cyclotron of 73 MeV at a dose of 60 Gy RBE (Radiation Biologic Equivalent) in 4 fractions. The dosimetry was based on the three-dimensional reconstruction of the eye and the tumor using EYE-PLAN software (Bebington, Wirral, United Kingdom).

From these 3,374 patients, we selected historical patients with choroidal melanoma of >10 mm in diameter and >5 mm in thickness. These inclusion criteria were established to obtain a sufficient number of treated patients and because in our cohort, these patients developed an NVG in ~50% at 5 years with a peak of incidence between 24 and 36 months (R.D.).

The first group, called P, was composed of patients treated with proton beam irradiation alone, and the second group, called PTTT, received proton beam irradiation combined with three sessions of transpupillary thermotherapy of the scar.¹⁰ These two groups were then compared with a third group, called PE, constituted of patients more recently treated with proton beam radiotherapy followed by an endoresection of the scar. We used endoresection after proton beam irradiation since >3 years. Correlation coefficient for age,

tumor diameter, and tumor thickness was calculated within the three groups. The study adhered to the tenets of the Declaration of Helsinki and was approved by of the Institutional Ethics Review Board.

Evaluated Data

Demographic data and tumor characteristics at baseline were recorded (i.e., tumor location, diameter, thickness, tumor nodes metastasis classification, initial presence of a retinal detachment, ciliary body invasion). Histologic and genomic analyses by array-comparative genomic hybridization were recorded only in the endoresection group.

Complications (i.e., exudative retinal detachment, tumor relapse, NVG, rate of secondary enucleation) were collected. Visual acuity was recorded at the first visit and at 1 year and 5 years of follow-up. For the endoresection group of patients, we have also collected the mean time between irradiation and surgery, intraoperative, and postoperative complications.

Data used in this paper were recorded in a database (Macro Infermed 3.075). Statistical analysis was done with the software R version 2.5.1 (<http://cran.r-project.org>).

Endoresection Technique

The technique was the same described by Damato et al⁹ modified by Bechrakis et al¹² with some differences because of the surgeon habits. Surgical procedure was performed by one surgeon (N.C.) trained to vitreoretinal surgery. A conventional (20 gauge) 3 port pars plana vitrectomy was performed with caution to the lens and complete posterior vitreous detachment was made. Special care was also taken to completely detach the hyaloid overlying the tumor. A special three functions probe including endoillumination, endodiathermy, and passive extrusion of fluid was used. After endodiathermy of the vessels around the tumor base, the endoresection of the tumor was started after application of 1 to 2 mL of perfluorocarbon liquid (PFCL) used to drain the subretinal fluid and to prevent subretinal hemorrhage at the posterior pole. The PFCL was regularly reinjected when the retina was totally detached to maintain the retina attached. Endoresection was done with regular endodiathermy of the bleeding vessels and increased intraocular pressure and decreased blood pressure when it was possible. The tumor was excised up to the sclera. Careful endodiathermy was applied around the coloboma after decreasing the intraocular pressure to check remaining bleeding. Under PFCL, endolaser was applied around the coloboma scar combined with cryoapplication of the anterior border. The remaining flat tumor residues

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were treated with laser. Perfluorocarbon liquid was then exchanged with standard silicon oil. After a mean period of 6 months, cataract extraction by phacoemulsification was realized with a posterior capsulorhexis and an anterior extraction of the silicone oil. A lens was then placed in the capsular bag.

Results

Of 3,374 patients treated for uveal melanoma at Institut Curie, 171 patients met inclusion criteria; 63 patients were treated with proton beam followed by an endoresection, 57 patients had proton beam irradiation alone, and 51 patients received proton beam irradiation combined with transpupillary thermotherapy.

The “endoresection” group (PE) was composed of 63 patients with a mean age at diagnosis of 57 years (range, 19–81 years), 65% were younger than 60 years, 42 were men (67%), and 21 were women (33%). Demographic data and tumor characteristics are summarized in Table 1. Mean largest basal tumor diameter was 13.9 mm (median, 14 mm; [range, 8–19 mm]). Tumors between 10 and 15 mm in diameter represented 59% of the patients. The mean tumor thickness was 8.6 mm (median, 8.7 mm range, 6–12.2 mm). Most of the patients (81%) had tumor thickness between 5 and 10 mm. Initial exudative retinal detachment was present in 79.4% of the cases, associated with an intravitreal hemorrhage in 4.8%. Initial cataract was present in 3 patients (4.75%), none of them was pseudophakic. Ciliary body invasion was

present in 5 patients (8%). Tumor extended to optic disk in 5 patients (8%). A preexistent glaucoma was noted in 1 patient (1.9%). The mean time between proton beam irradiation and endoresection was 3 months (range, 1–12 months).

Intraoperative or immediate postoperative complications (from D0 to D30) were noted in 17 patients (27%). The resection of the tumor was not possible in four cases. In two of the four cases, the tumoral tissue was too fibrotic to be removed by the vitrector handpiece. In two of the four cases, an inability to flatten a massive exudative retinal detachment (one case) and a huge bleeding arising from the ciliary body that was impossible to coagulate (one case) were prevented from performing the resection. Hyphema was noted in 10 patients, and preretinal hemorrhage was noted in 3 patients. We also observed one case of presumed endophthalmitis. Other minor complications were noted in five patients (i.e., inflammation, cataract because of lens trauma, silicone oil passage into the anterior chamber).

Histologic and genomic analyses of the endoresected tumor were done in 59 patients. Results showed that 72.5% of tumors were of epithelioid or mixed cell type, 11.8% were spindle cell tumors, and 13.7% were undetermined. Tumoral cells could not be found only in one case. The genomic status was evaluated by array-comparative genomic hybridization^{13,14} and showed that 25 tumors (62.5% of evaluated cases) had low-risk genomic profile (including 2 cases with partial 3q loss, gain of 6p, and lack of whole 8q gain)

Table 1. Baseline Patient Characteristics in the Three Groups

	PE (n = 63)	P (n = 57)	PTTT (n = 51)	P
Mean age, years	57 (range, 19–81)	56 (range, 22–64)	59 (range, 31–68)	0.29 (ns)
Gender				0.88 (ns)
Male, %	66.5	60	45	
Female, %	33.5	40	55	
Tumor site				0.58 (ns)
Posterior	25.5% (n = 16)	21% (n = 12)	30% (n = 15)	
Equator	73% (n = 46)	79% (n = 45)	70% (n = 36)	
Anterior	2% (n = 1)	0% (n = 0)	0% (n = 0)	
Median tumor diameter (range), mm	14 (8–19)	17.5 (10–23.3)	17.4 (12.5–22.7)	<0.001
Median tumor thickness (range), mm	9 (6–12.2)	8 (5.1–11.3)	8 (5.3–11.8)	0.004
Mean tumor volume (range), cm ³	1.07 (0.46–2.43)	1 (0.56–2.25)	0.97 (0.59–2.62)	0.87
TNM				<0.001
T1	0%	0%	0%	
T2	19% (n = 12)	2% (n = 1)	0%	
T3	75% (n = 47)	51% (n = 29)	63% (n = 32)	
T4	6% (n = 4)	47% (n = 27)	37% (n = 51)	
Initial RD				0.001
Present	80% (n = 50)	47% (n = 27)	51% (n = 26)	
Absent	20% (n = 13)	53% (n = 30)	49% (n = 25)	

Anterior, anterior to the equator; equator, within the equator; ns, nonsignificance; posterior, posterior to the equator; P, proton beam alone; PE, proton beam + endoresection; PTTT, proton beam + transpupillary thermotherapy; TNM, tumor nodes metastasis classification; RD, retinal detachment.

and 15 cases (37.5%) had high-risk profile, including one case with partial 3p loss and gain of whole 8q. Five cases (11% of the total samples) were not evaluated because of a balanced profile, presumably related to low tumor cell content.

In this PE group, the mean follow-up time was 23 months (range, 7–44 months). Patients were alive without metastasis in 89.5%, alive with metastasis in 7%, alive with a second cancer in 1.7%, died of metastasis in 1.7%, and lost to follow-up in 1.7%. All the tumors that developed metastasis (6 of 59 patients; 10%) were in the high-risk genomic group.

Table 2 summarizes visual acuity evolution at 2 years. Patients had decreased vision in 79.5%, vision was unchanged in 15.25%, improved in 3.4%, and data were not available in 1.5%. Long-term complications were cataract in 56% and retinal detachment requiring a second surgery in 5%. Neovascular glaucoma was noted in 4 cases (6.7%). Patients who did not develop NVG represented 96.5% at 1 year of follow-up and 94.3% at 2 years of follow-up. Patients who developed NVG were those in whom the resection of the tumor was not possible. Three patients underwent a secondary enucleation for complication not related to NVG, but for a massive hemorrhage associated with pain (one), and for a painful phthisis (two patients). The 2 phthisis were noted when endoresection was performed for tumors with a very large diameter (>16 mm).

Demographic data and tumor characteristics for the PE, P, and PTTT groups are summarized in Table 1. Evolution of visual acuity within these groups is shown in Table 3.

At 2 years, NVG occurred in 57.8% of the P patients ($n = 33$) and in 49% of the PTTT patients ($n = 25$). In contrast, NVG occurred in 6.7% ($n = 4$) of the PE patients. Secondary enucleation was necessary for NVG in 24.5% of patients in the P group and in 9.8% of patients in the PTTT group (Table 3).

At 5 years, metastasis occurred in 33.33% of the patients in the P group and in 37.25% of the patients in the PTTT group (Table 2).

Statistical comparison between the 3 groups was nonsignificant and was for age ($P = 0.29$), gender ($P = 0.88$), and tumor location ($P = 0.58$). We observed that tumor diameter was smaller in the PE group compared with the P and PTTT groups ($P < 0.001$). However, when the tumor volume was calculated, the difference within the 3 groups was nonsignificant ($P = 0.87$) (Table 1).

Survival without secondary enucleation was at 2 years (IC 95%) for PE 96.2%; for P 88.8% and for PTTT 98% with a $P = 0.203$. Survival without neovascular glaucoma at 2 years (IC 95%) was for PE=92.7% (85.1–1.00); for P=54.6% and for PTTT = 62.1% with a $P < 0.0001$. Differences between endoresection (PE) group and both proton (P) group and proton + TTT (PTTT) group in reducing the rate of neovascular glaucoma was statistically significant (Figure 1 and Table 2).

Relative risk of developing an NVG by tacking as reference the group treated with proton beam irradiation alone, relative risk = 1. The relative risk of the proton combined with transpupillary thermotherapy (PTTT) group was 0.79 (20% reduction of the risk). For the proton followed by endoresection (PE) group, the relative risk was 0.18 (82% reduction of the risk).

Discussion

In many countries, treatment of uveal melanoma is conservative in most cases. Radiotherapy using different modalities and different types of ionizing radiations are used.^{15,16} The choice of radiotherapy modalities depends on countries. In the United States and Europe, proton therapy and iodine 125 plaques are most commonly used.¹⁵ In France, the majority of patients are treated with proton beam radiotherapy and less frequently with iodine 125 plaque radiotherapy.¹⁷ Despite the type or modalities, the treatment of the largest tumors remains an ongoing challenge. Eye retention depends on different systemic (i.e., diabetes) or local (i.e., tumor involving the ciliary body,

Table 2. Outcome and development of complications in the 3 groups

Course of VA at 2 Years	PE (n = 63)	P (n = 56)	PTTT (n = 49)	P
Initial VA				0.67 (ns)
<1/100	17% (n = 11)	18% (n = 10)	10% (n = 5)	
1/100 to 20/50	59% (n = 36)	55% (n = 31)	70% (n = 34)	
>20/50	24% (n = 16)	27% (n = 15)	20% (n = 10)	
Final VA				0.54 (ns)
<1/100	68% (n = 42)	67% (n = 41)	72% (n = 31)	
1/100 to 20/50	13% (n = 18)	28% (n = 17)	26% (n = 11)	
>20/50	5% (n = 3)	5% (n = 3)	2% (n = 1)	

ns, Nonsignificance; P, proton beam alone; PE, proton + endoresection; PTTT, proton + transpupillary thermotherapy; VA, visual acuity % of patients.

Table 3. Course of visual acuity in the 3 groups

Outcome at 2 Years	PE (n = 63)	P (n = 57)	PTTT (n = 51)	P
NVG	7% (n = 4)	58% (n = 33)	49% (n = 25)	<0.0001
Secondary enucleation	6% (n = 3)	25% (n = 14)	10% (n = 5)	0.203
Metastasis	10% (n = 6)	14% (n = 8)	14% (n = 7)	ns
Median follow-up (range), months	23 (20–26)	112 (107–126)	99 (89–124)	

ns, Nonsignificance; NVG, % of patient with neovascular glaucoma; P, proton beam alone; PE, proton beam + endoresection; PTTT, proton beam + transpupillary thermotherapy.

thickness, location near the macula) factors.² For medium to large size choroidal melanomas, the risk of secondary enucleation because of NVG is near 30% at 5 years.¹⁸ Extra large tumors defined by a diameter over 20 mm or thickness over 10 mm require primary enucleation because radiotherapy will induce severe side effects, especially NVG in >30%.⁶

To increase eye retention rate, authors developed adjunctive therapy. Authors showed that proton beam irradiation followed by TTT decrease irradiation complications, especially exudative retinal detachment but had little effect on NVG.^{8,10} In our study, we compared proton beam followed by endoresection of the scar with proton beam alone and proton beam followed by TTT and we showed that endoresection provided better results in preventing NVG. Damato and Lecuona¹⁹ developed surgical techniques to remove the tumor before irradiation (exoresections or partial lamellar uveo sclerectomy) or by an internal way (endoresection) with good results in term of eye retention. We believe that exoresection is more challenging, particularly with large and posterior tumors, because of the need for hypotensive anesthesia and other measures to control intraoperative hemorrhage.^{20–22}

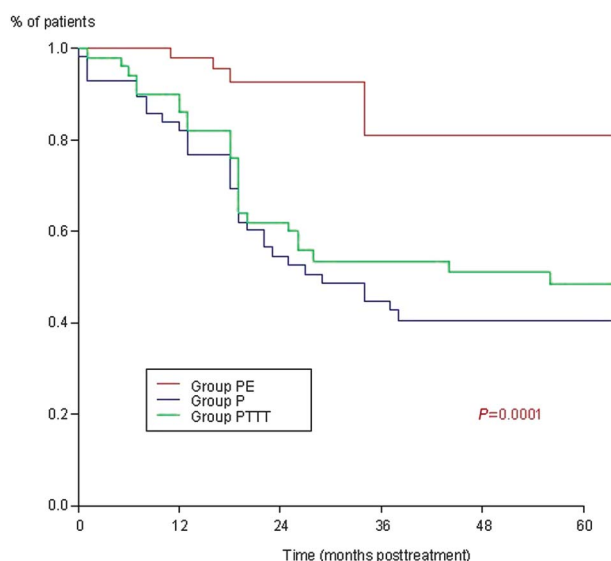


Fig. 1. Free interval without NVG.

Endoresection was first described by Peyman et al²³ in 1988 and developed by Damato et al.⁹ Endoresection was realized as a curative treatment by several teams.^{9,24} However, these studies showed that endoresection without previous irradiation can be associated with tumor recurrences that may increase the risk of metastasis.^{25–29} Bechrakis et al¹² showed that endoresection after tumor “devitalisation” by irradiation is effective in reducing the risk of tumor recurrences with good results in terms of eye retention. We can hypothesize that the prevention of NVG by removing the irradiated necrotic material can be explained by a “toxic” role of the necrotic scar. In our study, we showed that endoresection after proton beam is feasible. However, we performed secondary enucleation for three patients not because of an NVG but because a phthisis occurred. In these enucleated eyes, no tumor recurrence was found on histologic examination. This phthisis was probably because of a too large resection zone. Neovascular glaucoma was observed when tumor resection was not possible because the tumor tissue was too fibrotic or related to a huge exudative retinal detachment. In these patients, ocular pressure was controlled with topical treatment. With experience, we recommend endoresection in patients with a tumor diameter of <17 mm. Tumors that invade the ciliary body are not considered as a good candidate for endoresection and are probably best managed by cilio-choroidectomy.¹⁹ In our experience, a bleeding arising from the anterior tumor margin is quite impossible to coagulate by an internal way. Despite these complications, we showed that endoresection after proton beam reduced significantly the rate of NVG and probably the need of the secondary enucleation with better selection of patients compared with patients treated with proton beam alone or proton beam followed by TTT. Visual acuity was not better in the endoresection group than in the other groups. Endoresection after proton beam irradiation did not really prevent visual loss. Despite all, eye retention is very important for French patients who are reluctant to enucleation. We did not observe any local recurrences compared with studies in which patients were treated with endoresection alone.^{26–28} Our follow-up is probably too short to give definitive results on this

point. However, local recurrences described in these studies appeared in maximum of 3 years.^{26,27,30} Similarly, according to Gragoudas et al,³¹ local recurrences represent 4% of all the eyes treated with proton beam irradiation and the highest rate of failure was observed during the first year after therapy. We did not recommend endoresection without a previous irradiation of the tumor because studies showed that local recurrences increased significantly the risk of metastasis.^{31,32}

In our series, endoresected patients who have a high risk of developing metastasis are those with a large choroidal tumor, epithelioid or mixed cell type tumors that represent 73% of the cases, and high-risk genomic tumors for 38% of patients. It is worth to notice that postirradiation endoresection samples are in a large majority of cases, an adequate material for risk evaluation by genomic profiling. After 2 years of follow-up, 89.5% of the patients did not develop metastasis. However, our follow-up in this experimental group is not long enough to evaluate the real influence of the surgery on overall survival.

Endoresection is a part of multimodality treatment of choroidal melanoma. Further studies are needed to compare surgical adjuvant approaches to medical postirradiation intravitreal injections of anti-vascular endothelial growth factor associated or not with laser photocoagulation. This new anti-vascular endothelial growth factor adjuvant therapy, more and more used, also needs to be evaluated.

Conclusion

Endoresection associated with previous irradiation of choroidal melanoma is a safe technique that prevents NVG. Future studies are needed to specify the place of surgery (exoresection and endoresection) in the choroidal melanoma therapeutic arsenal.

Key words: choroidal melanoma, endoresection, proton beam radiotherapy, complication, local surgery.

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